

QUBO Formulation: Electrical Distribution Network Reconfiguration Problem

Electrical Distribution Network Reconfiguration Problem

Problem Statement. Given a set of N candidate radial switch configurations for an electrical distribution network, each configuration i is associated with a cost c_i that accounts for power-loss losses and any voltage or feasibility penalties. The objective is to select exactly one configuration such that the total network loss cost is minimized.

Decision Variables. We introduce binary decision variables $x_i \in \{0, 1\}$ for each candidate configuration $i \in \{0, \dots, N-1\}$. The variable x_i equals 1 if configuration i is selected, and 0 otherwise.

QUBO Derivation. Step 1. The original optimization problem is formulated as:

$$\min_{x \in \{0,1\}^N} \sum_{i=0}^{N-1} c_i x_i. \quad (1)$$

Step 2. The requirement to select exactly one configuration is expressed as the linear equality constraint:

$$\sum_{i=0}^{N-1} x_i = 1. \quad (2)$$

Step 3. We convert the constraint into a quadratic penalty term. Let $\lambda > 0$ be a penalty weight. The penalty function is:

$$P(x) = \lambda \left(\sum_{i=0}^{N-1} x_i - 1 \right)^2. \quad (3)$$

Expanding the square yields:

$$P(x) = \lambda \left(\sum_{i=0}^{N-1} x_i^2 + 2 \sum_{0 \leq i < j \leq N-1} x_i x_j - 2 \sum_{i=0}^{N-1} x_i + 1 \right). \quad (4)$$

Applying the idempotent property $x_i^2 = x_i$ for binary variables, we simplify the expression to:

$$P(x) = \lambda \left(- \sum_{i=0}^{N-1} x_i + 2 \sum_{0 \leq i < j \leq N-1} x_i x_j + 1 \right). \quad (5)$$

Step 4. Combining the objective and the penalty term gives the full QUBO Hamiltonian $H(x)$:

$$H(x) = \sum_{i=0}^{N-1} c_i x_i + \lambda \left(- \sum_{i=0}^{N-1} x_i + 2 \sum_{0 \leq i < j \leq N-1} x_i x_j + 1 \right). \quad (6)$$

Step 5. By collecting coefficients for each monomial, we read off the QUBO matrix entries and constant offset c :

$$Q_{i,i} = c_i - \lambda, \quad i \in \{0, \dots, N-1\}, \quad (7)$$

$$Q_{i,j} = 2\lambda, \quad 0 \leq i < j \leq N-1, \quad (8)$$

$$c = \lambda. \quad (9)$$

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that any violation of the exactly-one constraint results in an energy strictly greater than the minimum achievable objective value. The maximum possible reduction in the objective cost from selecting a configuration is bounded by $\max_i c_i$. Therefore, choosing $\lambda > \max_i c_i$ guarantees that the penalty incurred by any infeasible state (where $\sum_{i=0}^{N-1} x_i \neq 1$) dominates any potential gain in the objective function, thereby enforcing feasibility at the global minimum.

Correctness Argument. Minimizing $H(x)$ over $\{0, 1\}^N$ recovers the optimal configuration because feasible solutions satisfy $\sum_{i=0}^{N-1} x_i = 1$, causing the penalty term to vanish. For these solutions, $H(x)$ equals the original objective $\sum_{i=0}^{N-1} c_i x_i$. Conversely, any infeasible solution violates the constraint, incurring a penalty of at least λ . Since λ exceeds the maximum possible objective value, every infeasible state has strictly higher energy than the best feasible state. Consequently, the global minimum of $H(x)$ must correspond to a feasible configuration that minimizes the original cost.

Illustrative Example. Consider a small instance with $N = 3$ candidate configurations and costs c_0, c_1, c_2 . Setting $\lambda = 100$, the Hamiltonian becomes:

$$H(x) = c_0 x_0 + c_1 x_1 + c_2 x_2 + 100(x_0 + x_1 + x_2 - 1)^2 \quad (10)$$

$$= c_0 x_0 + c_1 x_1 + c_2 x_2 + 100(-x_0 - x_1 - x_2 + 2x_0 x_1 + 2x_0 x_2 + 2x_1 x_2 + 1). \quad (11)$$

The corresponding QUBO parameters are $Q_{0,0} = c_0 - 100$, $Q_{1,1} = c_1 - 100$, $Q_{2,2} = c_2 - 100$, $Q_{i,j} = 200$ for all $i \neq j$, and offset $c = 100$. Evaluating $H(x)$ over all $2^3 = 8$ binary

assignments shows that the three feasible states $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ yield energies c_0 , c_1 , and c_2 , respectively. The infeasible states incur an additional penalty of 100, 200, or 300. Thus, the global minimum occurs at the configuration x^* with $x_k^* = 1$ for $k = \arg \min_i c_i$, and the minimum energy equals $\min_i c_i$.